

YONAGUNIJIMA, Japan

WMO: 479120

Lat: 24.4667N

Lon: 123.0106E

Elev: 120

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.2	56.9	41.3	38.4	57.5	43.5	41.8	59.1	36.6	59.5	34.4	60.3	24.2	10	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	89.6	80.9	88.7	80.6	87.8	80.3	82.5	87.1	81.8	86.5	81.1	85.9	11.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.1	162.2	85.5	80.5	158.8	85.1	79.8	155.4	84.8	46.5	87.4	45.7	86.6	45.1	86.1	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.7	31.4	28.6	52.5	91.2	2.6	1.3	50.6	92.2	49.1	92.9	47.6	93.7	45.7	94.6
			46.3	83.2	2.4	1.2	44.5	84.0	43.1	84.7	41.8	85.4	40.1	86.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.5	65.5	66.4	68.9	73.7	78.0	82.3	84.3	83.8	81.7	78.2	73.9	68.3
	DBStd	7.54	4.48	5.05	4.77	4.20	3.33	2.75	1.70	1.74	2.34	2.72	3.44	4.44
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	136	50	42	20	2	0	0	0	0	0	1	21	
	CDD50	9295	480	459	586	711	868	969	1063	1048	951	876	717	567
	CDD65	3957	65	81	142	263	403	519	598	583	501	411	268	123
	CDH74	34628	29	79	289	1185	3095	5814	7461	7078	5382	3089	986	141
	CDH80	10180	0	0	3	81	493	1883	3057	2705	1508	398	51	1
Wind	WSAvg	14.9	17.8	17.1	15.5	13.8	11.9	12.3	12.3	11.9	13.6	17.0	17.1	18.3
Precipitation	PrecAvg	91.4	7.8	6.2	6.2	6.0	7.5	6.6	5.2	8.5	11.0	9.8	8.4	7.6
	PrecMax	171.7	17.1	12.6	15.5	13.1	23.3	15.2	16.9	20.9	43.7	43.5	21.1	20.2
	PrecMin	46.2	0.3	2.4	1.8	2.5	1.0	1.3	0.4	0.7	1.1	1.4	1.8	1.3
	PrecStd	24.8	3.7	2.7	3.0	2.7	4.7	3.4	3.9	5.8	8.8	8.7	4.3	4.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.6	78.5	80.3	83.9	87.2	89.3	90.8	90.8	89.7	86.8	83.4	79.1
		MCWB	70.1	72.5	74.4	77.9	80.4	81.6	81.2	80.9	80.9	79.3	76.7	72.7
	2%	DB	74.6	76.2	78.4	82.2	85.3	88.1	89.7	89.5	87.8	84.9	81.5	77.1
		MCWB	68.5	70.6	72.9	76.3	79.1	81.0	80.9	80.7	80.0	78.1	75.1	71.0
	5%	DB	73.0	74.7	77.0	80.7	84.0	87.2	88.8	88.4	86.7	83.5	80.0	75.6
		MCWB	67.2	69.6	71.8	75.6	78.2	80.6	80.6	80.4	79.3	76.9	74.0	69.5
	10%	DB	71.6	73.4	75.6	79.3	82.7	86.2	87.9	87.3	85.7	82.2	78.6	74.1
		MCWB	66.1	68.3	70.5	74.7	77.4	80.2	80.2	80.1	78.7	75.7	73.0	68.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.2	73.8	75.8	79.0	81.1	82.8	83.4	83.5	82.1	80.7	77.8	74.7
		MCDB	74.9	77.0	78.8	82.4	85.4	87.3	88.3	87.4	87.6	85.1	81.8	77.6
	2%	WB	70.2	72.2	74.2	77.4	80.0	81.9	82.4	82.3	81.0	79.3	76.6	72.6
		MCDB	73.1	75.3	77.5	80.9	84.0	86.4	87.1	86.9	86.1	83.5	80.3	75.8
	5%	WB	68.7	70.7	72.7	76.4	79.1	81.3	81.6	81.6	80.3	78.2	75.4	71.0
		MCDB	71.9	74.0	76.4	79.9	82.8	85.7	86.6	86.3	85.1	82.3	78.8	74.3
	10%	WB	67.2	69.1	71.3	75.3	78.3	80.7	81.0	80.9	79.7	77.1	74.2	69.4
		MCDB	70.9	72.6	75.0	78.8	81.9	85.1	86.2	85.7	84.3	81.3	77.8	73.2
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.6	7.2	6.9	7.0	6.4	7.3	7.5	7.2	6.2	6.0	5.9
		MCDBR	7.9	8.1	8.4	7.4	7.6	6.7	8.1	8.4	8.1	7.7	7.4	7.5
		MCWBR	5.7	5.9	6.2	5.0	4.3	3.5	3.6	3.9	4.2	4.4	4.7	5.2
	5% WB	MCDBR	7.0	7.4	8.0	7.1	7.2	6.6	7.6	7.9	7.8	7.2	6.8	6.9
		MCWBR	5.9	6.3	6.8	5.2	4.4	3.7	3.8	4.0	4.2	4.4	4.9	5.6
Clear-Sky Solar Irradiance	taub	0.519	0.541	0.632	0.638	0.557	0.509	0.487	0.507	0.508	0.517	0.514	0.524	
	taud	1.975	1.953	1.763	1.804	2.051	2.213	2.284	2.211	2.180	2.092	2.068	1.977	
	Ebn at Noon	230	236	222	224	242	252	258	253	250	240	230	222	
	Edh at Noon	52	56	71	70	54	46	43	46	46	49	47	50	
All-Sky Solar Radiation	RadAvg	646	803	1021	1230	1518	1763	2030	1836	1580	1203	816	643	
	RadStd	119	160	158	179	207	159	127	146	154	148	103	106	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.68	+0.56	+0.63	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	+0.49	N/A	N/A	+0.67	+0.47	+0.56	N/A	N/A	+175	+155

Nomenclature: See separate page